

Lecture 6: Selecting an Estimator, Model Fit, and Testing Assumptions

Feb 27, 2025

Review

Homework 2

- **Problem 2 of HW#2 are related to Lecture 2**

Empirical growth plot

- page 25 of book

Non-parametric

- page 27 of book

Parametric (fitted OLS trajectories superimposed on individual empirical growth plots)

- page 32 of book

OLS trajectories across participants (OLS regressions conducted and visualized with the mean trajectory line)

- page 34 and 38 of book

From Last Lecture...

- Fitting the maximum likelihood estimator the linear mixed-effects model (LME) through the ball weight problems
- Fitting the restricted maximum likelihood estimator for the variance parameters in the LME model

Random Effects

- How to estimate the random effects
- Likelihood function via conditional densities

High-level Ideas: Mathematical

- Data: $(Y_{ij}, x_{ij1}, \dots, x_{ijp}), i = 1, \dots, m, j = 1, \dots, n_i$

- (A Simple LME) Model:

$$Y_{ij} = \beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + \mathbf{b}_i + \epsilon_{ij}$$

where random effects $b_i \sim N(0, \sigma_b^2)$, independent of $\epsilon_{ij} \sim N(0, \sigma^2)$

- MLE: find $(\beta_1, \dots, \beta_p, \sigma_b, \sigma)$ that maximizes the likelihood function

$$L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) = \prod_{i=1}^m f(Y_{i1}, \dots, Y_{i,n_i})$$

- REML: find (σ_b, σ) that maximizes the restricted likelihood

$$L(\sigma_b, \sigma) = \iiint L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) d\beta_1 \dots d\beta_p$$

High-level Ideas: Computing

- **Data:** $(Y_{ij}, x_{ij1}, \dots, x_{ijp}), i = 1, \dots, m, j = 1, \dots, n_i$
- **Model:** $Y_{ij} = \beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + b_i + \epsilon_{ij}$ where $b_i \sim N(0, \sigma_b^2)$, independent of $\epsilon_{ij} \sim N(0, \sigma^2)$
- **MLE:** Given $\theta = \frac{\sigma}{\sigma_b}$, find $(\beta_1, \dots, \beta_p)$ and $(b_1, \dots, b_m)$ that minimizes

$$RSS = \sum_{i=1}^m \left(\sum_{j=1}^{n_i} \epsilon_{ij}^2 + \theta^2 b_i^2 \right) = \sum_{i=1}^m \left(\sum_{j=1}^{n_i} (Y_{ij} - (\beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + b_i))^2 + \theta^2 b_i^2 \right)$$

- This is equivalent to run the standard linear regression on $n_1 + \dots + n_m + m$ new data points:

$$\begin{pmatrix} Y_{ij}, x_{ij1}, \dots, x_{ijp}, 0, \dots, 1, \dots \\ (0, 0, \dots, 0, 0, \dots, \theta, \dots) \end{pmatrix} \text{ for } i = 1, \dots, m, j = 1, \dots, n_i$$

- **REML:** $\hat{\sigma}^2 = \frac{RSS}{n-p}$

Ball Weight Problems

- Two NYU students, “Jack” and “Kathy”, work together to find the weights of two balls, A and B, using a scale with random measurement errors. Each student took two measurements, and
- “Jack” measures ball “A” and “B” each one time, and the observed weights are 2 and 4 lbs.
- “Kathy” measures “B” and “A+B”, and observed weights are 3 and 5 lbs.

Questions: use the linear mixed effect model to find the weights of balls A and B.

Data → Model

	A	B	A+B
Jack	2	4	--
Kathy	--	3	5

- **Model: Observed Data = True Value + Noise** where
Noise = Subject-Random-Effects + Measurement Error

$$2 = Y_{11} = A + b_1 + \epsilon_{11}$$

$$4 = Y_{12} = B + b_1 + \epsilon_{12}$$

$$3 = Y_{21} = B + b_2 + \epsilon_{21}$$

$$5 = Y_{22} = A + B + b_2 + \epsilon_{22}$$

R code

```
data1 <- rbind( c(1,2,1,0), c(1,4,0,1), c(2,3, 0,1), c(2,5,1,1) );  
colnames(data1) <- c("ID", "Y", "X1", "X2");  
data1 = data.frame(data1);  
data1lme1 <- lmer( Y ~ X1 + X2 - 1 + (1 | ID), data= data1);  
summary(lme1)
```

```
> summary(lme1)
```

```
Linear mixed model fit by REML ['lmerMod']
```

```
Formula: Y ~ X1 + X2 - 1 + (1 | ID)
```

```
Data: data1
```

```
REML criterion at convergence: 4.1
```

```
Scaled residuals:
```

```
   Min    1Q  Median    3Q   Max
-0.6708 -0.3354  0.0000  0.2795  0.4472
```

```
Random effects:
```

```
Groups  Name      Variance Std.Dev.
ID      (Intercept) 0.25    0.5000
Residual                0.05    0.2236
```

```
Number of obs: 4, groups: ID, 2
```

```
Fixed effects:
```

```
Estimate Std. Error t value
X1  1.8000    0.2345   7.675
X2  3.6500    0.2872  12.708
```

```
Correlation of Fixed Effects:
```

```
 X1
X2 0.408
```

R code/output

```
> ## Fixed-effect parameter
```

```
> fixef(lme1)
```

```
 X1  X2
1.80 3.65
```

Random-Effects

> ### Estimating the random effects

> **ranef(lme1)**

\$ID

(Intercept)

1 0.25

2 -0.50

$\hat{b}_1 = 0.25$

$\hat{b}_2 = -0.50$

with conditional variances for “ID”

R codes & Output (cont.)

```
> ##
```

```
> coef(lme1)
```

```
$ID
```

```
  (Intercept) X1  X2
```

```
1      0.25 1.8 3.65
```

```
2     -0.50 1.8 3.65
```

```
attr("class")
```

```
[1] "coef.mer"
```

$$2 = Y_{11} = A + b_1 + \epsilon_{11} \quad (\hat{\epsilon}_{11} = -0.05)$$

$$4 = Y_{12} = B + b_1 + \epsilon_{12} \quad (\hat{\epsilon}_{12} = 0.10)$$

$$3 = Y_{21} = B + b_2 + \epsilon_{21} \quad (\hat{\epsilon}_{21} = -0.15)$$

$$5 = Y_{22} = A + B + b_2 + \epsilon_{22} \quad (\hat{\epsilon}_{22} = 0.05)$$

Theoretical: Efficient Computing

- Models

$$\begin{aligned}2 &= Y_{11} = A + b_1 + \epsilon_{11} \\4 &= Y_{12} = B + b_1 + \epsilon_{12} \\3 &= Y_{21} = B + b_2 + \epsilon_{21} \\5 &= Y_{22} = A + B + b_2 + \epsilon_{22}\end{aligned}$$

- Objective Function

$$\begin{aligned}RSS &= (2 - A - b_1)^2 + (4 - B - b_1)^2 + \theta^2 b_1^2 \\&\quad + (3 - B - b_2)^2 + (5 - A - B - b_2)^2 + \theta^2 b_2^2\end{aligned}$$

- Linear regression on new data

(2, 1, 0, 1, 0)
(4, 0, 1, 1, 0)
(0, 0, 0, theta, 0)
(3, 0, 1, 0, 1)
(5, 1, 1, 0, 1)
(0, 0, 0, 0, theta)

- Given $\theta = \sqrt{\frac{0.05}{0.25}} = \sqrt{\frac{1}{5}}$ to get the results

R code for linear regression

```
theta = sqrt(0.05/0.25)
## new data
data2 <- rbind( c(2,1,0, 1,0),  c(4,0,1, 1,0), c(0,0,0,theta,0),
c(3,0,1, 0, 1),  c(5,1,1, 0, 1), c(0,0,0,0, theta) );

colnames(data2) <- c( "Y", "A", "B", "b1", "b2" );
data2 <- data.frame(data2)

# standard regression on new data
lm(Y ~ . -1, data=data2)
```

Output

```
> lm(Y ~ . -1, data=data2)
```

Call:

```
lm(formula = Y ~ . - 1, data = data2)
```

Coefficients:

A	B	b1	b2
1.80	3.65	0.25	-0.50

R codes & Output (cont.)

```
> ##
```

```
> coef(lme1)
```

```
$ID
```

```
(Intercept) X1 X2
```

```
1      0.25 1.8 3.65
```

```
2     -0.50 1.8 3.65
```

```
attr("class")
```

```
[1] "coef.mer"
```

$$2 = Y_{11} = A + b_1 + \epsilon_{11} \quad (\hat{\epsilon}_{11} = -0.05)$$

$$4 = Y_{12} = B + b_1 + \epsilon_{12} \quad (\hat{\epsilon}_{12} = 0.10)$$

$$3 = Y_{21} = B + b_2 + \epsilon_{21} \quad (\hat{\epsilon}_{21} = -0.15)$$

$$5 = Y_{22} = A + B + b_2 + \epsilon_{22} \quad (\hat{\epsilon}_{22} = 0.05)$$

$$\begin{aligned} RSS &= \hat{\epsilon}_{11}^2 + \hat{\epsilon}_{12}^2 + \theta^2 \hat{b}_1^2 + \hat{\epsilon}_{21}^2 + \hat{\epsilon}_{22}^2 + \theta^2 \hat{b}_2^2 \\ &= 0.1 \quad \text{when } \theta = \frac{\hat{\sigma}}{\hat{\sigma}_b} = \sqrt{\frac{0.05}{0.25}} = \sqrt{\frac{1}{5}} \end{aligned}$$

$$\hat{\sigma}^2 = \frac{RSS}{n - p} = \frac{0.1}{4 - 2} = 0.05$$

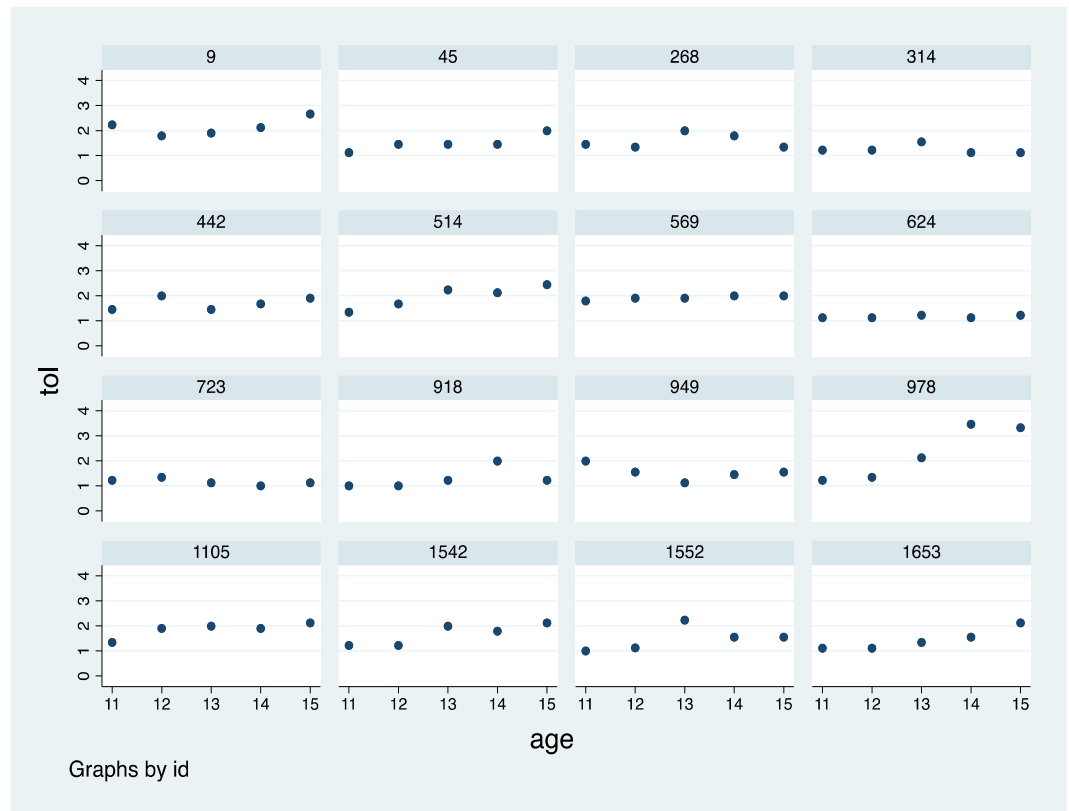
This Lecture

- Step 1: Fitting the unconditional models
 - Unconditional mean model
 - Unconditional growth model
- Step 2: Model building to the “final model”
 - Predictor only model
 - Predictor with control variable model
 - “Final” model

Person-Period ("long") data format

(Singer and Willet page 25)

```
. graph twoway scatter tol age, by(id) ylabel(0(1)4) xlabel(11(1)15)
```



Graphs by id

- Descriptive analyses

- How do people change over time?
- Is there growth?
- What is the functional form?

- Empirical growth plot

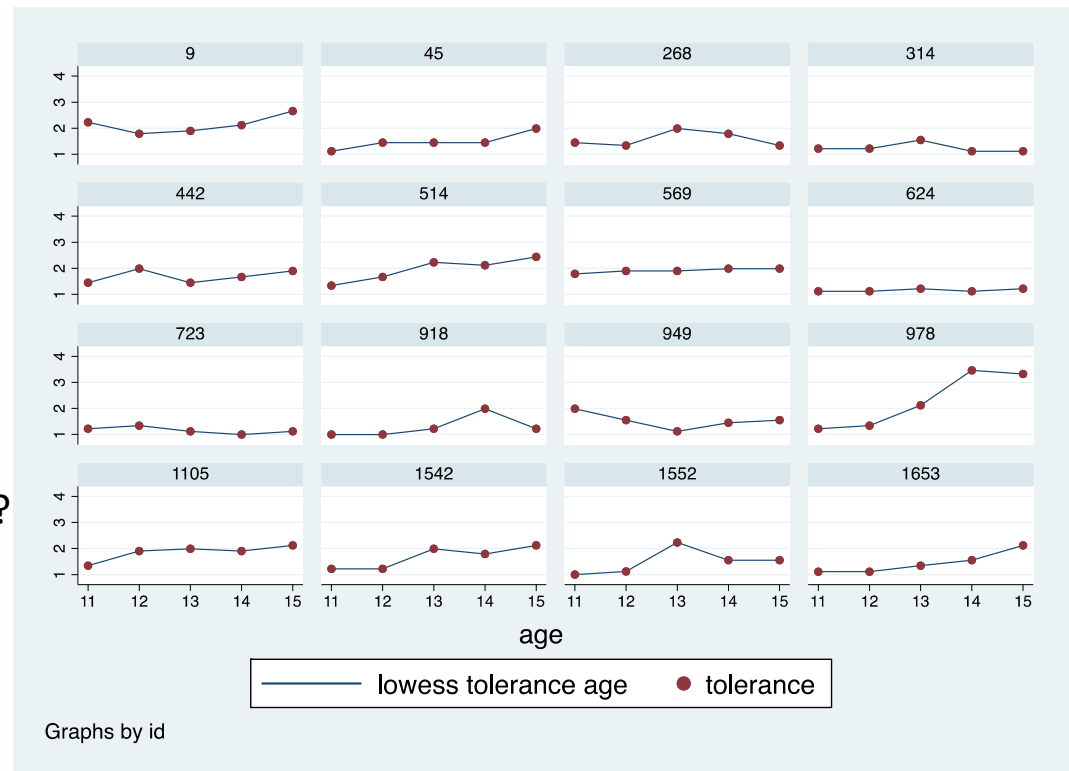
- Who is increasing, who is decreasing, who is increasing the most?
- What information can you garner from this plot?

Person-Period ("long") data format

(Singer and Willet pages 26 and 27)

- Non-parametric standardized approach
 - No functional form
 - “Let the data speak for themselves”
- What do you notice about the data?
 - Is the trajectory a bit clearer now?

. graph twoway (lowess tolerance age)(scatter tolerance age), by(id)

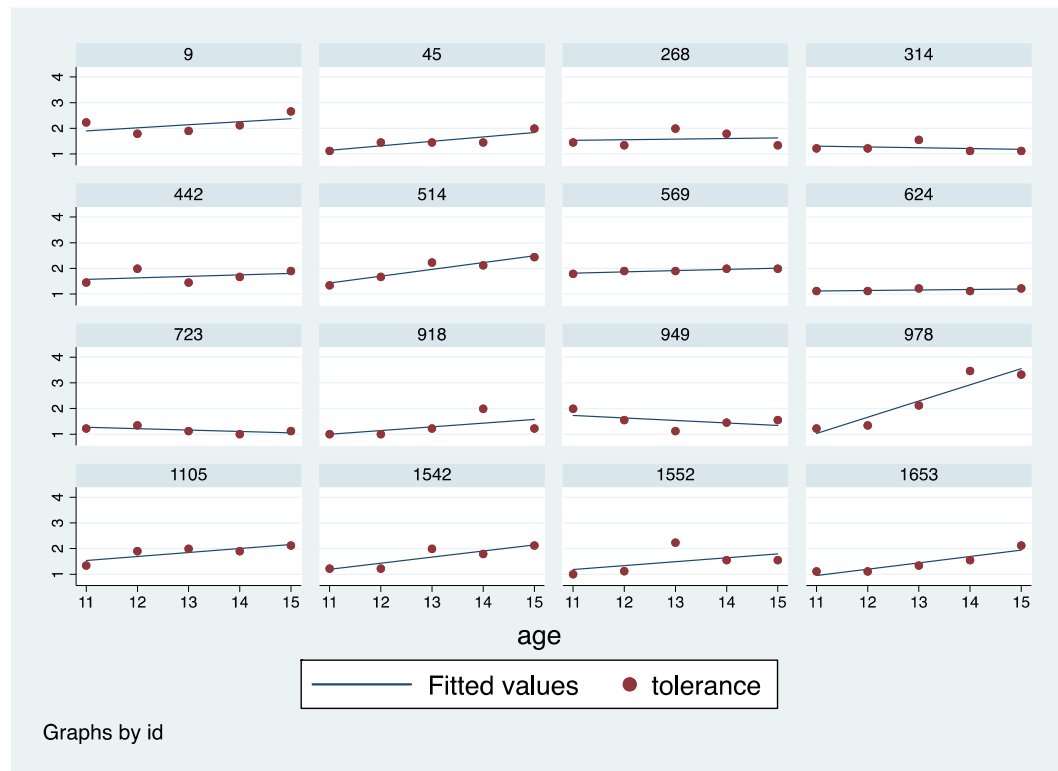


Person-Period ("long") data format

(Singer and Willet pages 32-34)

- Parametric standardized approach
 - OLS linear regression
 - Fitted OLS trajectories superimposed on empirical growth plots for participants
- What do you notice about the data?
 - Anything different?
 - Confirmatory?

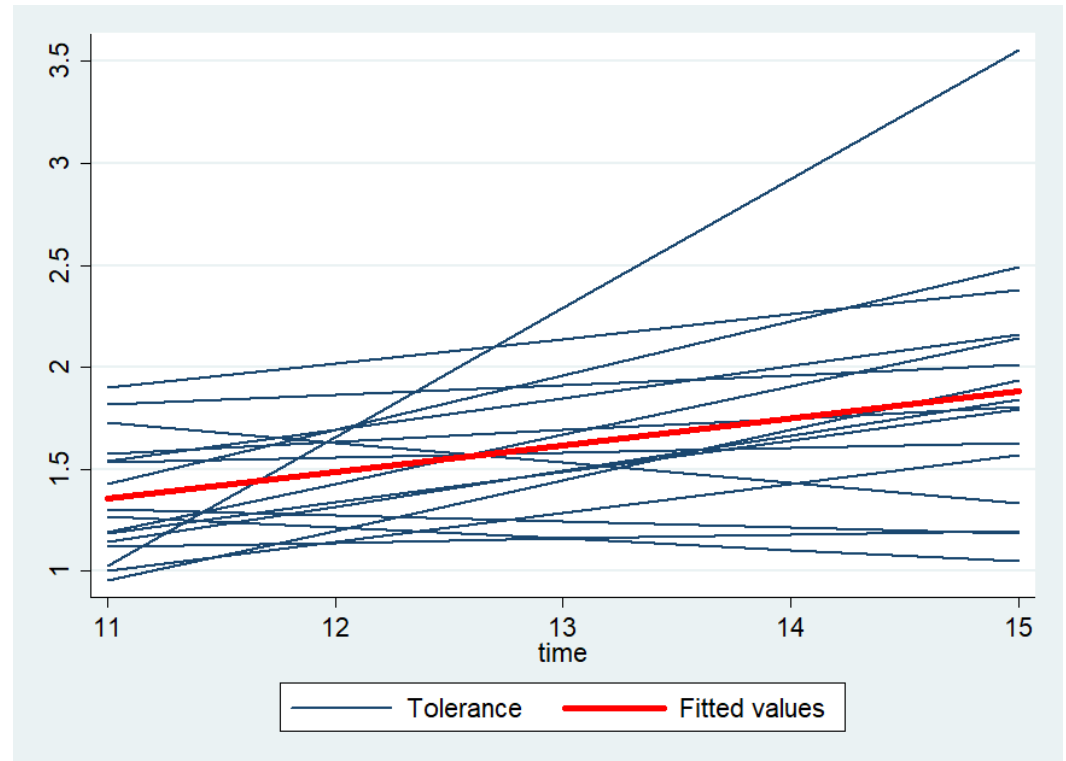
. graph twoway (lfit tolerance age)(scatter tolerance age), by(id)



Person-Period ("long") data format

(see dofile in Lecture 2 folder, Singer and Willet pages 34 and 38)

- Parametric standardized approach
 - OLS linear regression
 - Fitted OLS trajectories superimposed on empirical growth plots for participants
 - **Overall mean trajectory line**
- What do you notice about the data?
 - Anything different?
 - Confirmatory?



Unconditional mean model

$$Y_{ij} = \pi_{0i} + \varepsilon_{ij} \quad \text{OR} \quad Y_{ij} = \gamma_{00} + \zeta_{0i} + \epsilon_{ij}$$
$$\pi_{0i} = \gamma_{00} + \zeta_{0i}$$

- γ_{00} = grand mean (fixed effect)
 - ζ_{0i} = the amount person i's mean deviates from the population mean (between-person)
 - ϵ_{ij} = the amount the response on occasion j deviates from person i's mean (within-person)
 - $\epsilon_{ij} \sim N(0, \sigma_{\epsilon}^2)$
 - $\zeta_{0i} \sim N(0, \sigma_0^2)$
- ζ_{0i} = the distance between person i's mean and the population average mean
- The unconditional means represent the observed value of **Y** for individual **i** on occasion **j**, **which** is composed of deviations about the person-specific and the grand means

Unconditional growth model

$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \epsilon_{ij}$$

$$\pi_{0i} = \gamma_{00} + \zeta_{0i}$$

$$\pi_{1i} = \gamma_{10} + \zeta_{1i},$$

$$Y_{ij} = \gamma_{00} + \gamma_{10} * time_{ij} + \zeta_{0i} + \zeta_{1i} * time_{ij} + \epsilon_{ij}$$

- γ_{00} = average intercept (fixed effect)
- γ_{10} = average slope (fixed effect)
- ζ_{0i} = the amount person i's intercept deviates from the population intercept
- ζ_{1i} = the amount person i's slope deviates from the population slope
- ϵ_{ij} = the amount the response on occasion j deviates from person i's true change

trajectory

- $\epsilon_{ij} \sim N(0, \sigma_{\epsilon}^2)$

σ_{ϵ}^2 Summarizes the scatter of each persons data around his or her own linear change trajectory

- $\zeta_{0i} \sim N(0, \sigma_0^2)$

σ_0^2 Summarizes the between person variability

- $\zeta_{1i} \sim N(0, \sigma_1^2)$

σ_1^2 Summarizes the within person variability

- ζ_{0i} and ζ_{1i} have covariance σ_1^2

Learning Objectives

- Clarifications
 - Unconditional mean model
 - Descriptive Statistics
- Assessing model fit
 - AIC/BIC
 - Loglikelihood
 - Deviance
- Testing model assumptions
 - Checking functional form
 - Checking normality
 - Checking homoscedasticity
 - Testing model assumptions
- Group project review

Assessing Model Fit

Assessing Model Fit

- Loglikelihood
 - Just like in regression, we want to see the overall value of the LL increase (if negative get closer to 0) as we move from our mean only model to our “final” model
- Deviance statistic
 - Difference between the current model and saturated model (exactly the same as the residual sum of squares in linear regression model)
- AIC/BIC
 - Just like in regression, we examine the absolute value of both. When running several nested models, we would like to see a decrease in the AIC/BIC

Loglikelihood

- When the absolute value of the $-2LL$ decreases this is an indication of a better model fit
- Lets take a look at the Cook et al paper from last weeks class exercise
 - Which is the “best model” according to the data?
- The loglikelihood will always increase (if negative, decrease or get closer to zero) when additional variables are added
 - This is why we examine multiple fit indicators in tandem

Deviance

(Singer and Willett, pgs. 116-120)

(4.15)

$$\text{Deviance} = -2[LL_{\text{current model}} - LL_{\text{saturated model}}]$$

$$\text{Deviance} = -2LL_{\text{current model}}$$

- Deviance statistic compares the loglikelihood of the current and saturated model
- How much worse is the current model than the best model (the saturated model)
- A small deviance means the current model is a close fit with the saturated model. This indicates a nice model fit
- A large deviance mean the current model is a poor fit

Deviance

- To compare models....
 - One must be nested in the other
 - The same data must be used
 - The same number of cases must be used in both models
 - If FML comparing entire models
 - If REML only comparing the random effects

Deviance Example

- Mean model only
- -2LL = -335.07799

Per the Deviance formula we multiple by 2, which = **670.16**

Mixed-effects ML regression		Number of obs	=	246
Group variable: id		Number of groups	=	82
		Obs per group:		
		min	=	3
		avg	=	3.0
		max	=	3
		Wald chi2(0)	=	.
		Prob > chi2	=	.
Log likelihood = -335.07799				

Mixed-effects ML regression		Number of obs	=	246
Group variable: id		Number of groups	=	82
		Obs per group:		
		min	=	3
		avg	=	3.0
		max	=	3
Random-effects Parameters		Wald chi2(1)	=	20.11
		Prob > chi2	=	0.0000
Log likelihood = -318.83122				

- Growth model
- -2LL = -318.83122
- Per the Deviance formula we multiple by 2, which = **636.61**

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
id: Independent					
		var(age_14)	.1172599	.0446745	.0555715 .2474266
		var(_cons)	.5432039	.1170779	.3560423 .8287513
		var(Residual)	.3629466	.0533485	.272099 .4841261
LR test vs. linear model: chi2(2) = 78.65					
Prob > chi2 = 0.0000					

Deviance Example

$$670.16 - 636.61 = 33.5$$

33.5 > 16.27 (the .001 critical value for the χ^2 distribution)

We conclude that the unconditional growth model provides a better fit than the mean only model

Mixed-effects ML regression					Number of obs	=	246
Group variable: id					Number of groups	=	82
					Obs per group:		
					min	=	3
					avg	=	3.0
					max	=	3
Log likelihood = -335.07799					Wald chi2(0)	=	.
					Prob > chi2	=	.
Mixed-effects ML regression					Number of obs	=	246
Group variable: id					Number of groups	=	82
					Obs per group:		
					min	=	3
					avg	=	3.0
					max	=	3
Log likelihood = -318.83122					Wald chi2(1)	=	20.11
					Prob > chi2	=	0.0000
alcuse	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]		
age_14	.2706514	.060358	4.48	0.000	.1523519	.3889509	
_cons	.6513035	.1015525	6.41	0.000	.4522642	.8503428	
Random-effects Parameters					Estimate	Std. Err.	[95% Conf. Interval]
id: Independent							
var(age_14)					.1172599	.0446745	.0555715 .2474266
var(_cons)					.5432039	.1170779	.3560423 .8287513
var(Residual)					.3629466	.0533485	.272099 .4841261
LR test vs. linear model: chi2(2) = 78.65					Prob > chi2 = 0.0000		

AIC/BIC

(Singer and Willett, pgs. 120-122)

- Akaike Information Criterion (AIC)/ Bayesian Information Criterion (BIC)
 - Correct for some of the limitations of singularly using the loglikelihood and the Deviance statistic
- AIC controls for the increase in model parameters
- BIC controls for the number of parameters and sample size
- These models do not need to be nested as long as the same data is utilized
- The smaller the AIC/BIC the better the model

AIC/BIC

(Singer and Willett, pgs. 120-122)

- Under MLE, can use both fixed and random effect parameters
- Under REML only random effects
- Art and science
 - Not clear what is a significant drop
 - Should always be used together
 - Conceptual model important

Notes on Model Fit

- In Stata you can use the code **estat ic** to get some of the model fit statistics we just discussed
- Model fit statistics should be taken together. When all are trending in a specific direction, then you have more confidence in your model fit
- There are many more model fit techniques that we will discuss throughout the rest of the course, but these are the basics

A final comment about estimation and hypothesis testing

Two most common methods of estimation

Maximum likelihood (ML):

Seeks those parameter estimates that maximize the likelihood function, which assess the joint probability of simultaneously observing all the sample data actually obtained (implemented, e.g., in HLM and SAS Proc Mixed).

Generalized Least Squares (GLS): (& Iterative

GLS): Iteratively seeks those parameter estimates that minimize the sum of squared residuals (allowing them to be autocorrelated and heteroscedastic) (implemented, e.g., in MLwiN and stata xtreg).

A more important distinction: Full vs. Restricted (ML or GLS)

Full: Simultaneously estimate the fixed effects and the variance components.

- *Default in MLwiN & HLM*

Restricted: Sequentially estimate the fixed effects and then the variance components

- *Default in SAS Proc Mixed & stata xtmixed*

*Goodness of fit statistics apply to the entire model
(both fixed and random effects)
This is the method we've used in both the examples shown so far*

*Goodness of fit statistics apply to only the random effects
So we can only test hypotheses about VCs (and the models being compared must have identical fixed effects)*

Testing Model Assumptions

Testing Model Assumptions

(Singer and Willett, pgs. 127-137; Table 4.3; pg. 129)

- Checking functional form
 - Is the form linear, binomial, cubic, etc.
- Checking normality
 - Normality of the residual is key to many of the assumptions used for the random effects and for MLE
- Checking homoscedasticity
 - Key assumption of the random effects

Table 4.3: Strategies for checking assumptions in the multilevel model for change, illustrated using Model F of tables 4.1 and 4.2 for the alcohol use data

Assumption and what to expect is tenable	level-1 residual, ϵ_{ij}	level-2 residual, ζ_{0i}	level-2 residual, ζ_{1i}
<i>Shape.</i> Linear individual change trajectories and linear relationships between individual growth parameters and level-2 predictors.	Empirical growth plots suggest that most adolescents experience linear change with age. For others, the small number of waves of data (3) makes it difficult to declare curvilinearity making the linear trajectory a reasonable approximation.	Because <i>COA</i> is dichotomous, there is no linearity assumption for π_{0i} . With the exception of two extreme data points, the plot of π_{0i} vs. <i>PEER</i> suggests a strong linear relationship.	Because <i>COA</i> is dichotomous, there is no linearity assumption for π_{1i} . Plot of π_{1i} vs. <i>PEER</i> suggests a weak linear relationship.
<i>Normality.</i> All residuals, at both level-1 and level-2, will be normally distributed.	A plot of ϵ_{ij} vs. normal scores suggests normality. We find further support for normality in a plot of standardized ϵ_{ij} vs <i>ID</i> , which reveals no unusual data points.	A plot of ζ_{0i} vs. normal scores suggests normality. So does a plot of standardized ζ_{0i} vs. <i>ID</i> , which reveals no unusual data points. There is slight evidence of a floor effect in the outcome.	A plot of ζ_{1i} vs. normal scores suggests normality, at least in the upper tail. The lower tail seems compressed. We find further support for this claim when we find no unusual data points in a plot of standardized ζ_{1i} vs. <i>ID</i> . There is also evidence of a floor effect in the outcome.
<i>Homoscedasticity.</i> Equal variances of the level-1 and level-2 residuals at each level of every predictor.	A plot of ϵ_{ij} vs. AGE suggests approximately equal variability at ages 14, 15, and 16.	A plot of ζ_{0i} vs. <i>COA</i> suggests homoscedasticity at both values of <i>COA</i> . So does a plot vs. <i>PEER</i> , at least for values up to, and including, 2. Beyond this, there are too few cases to judge.	A plot of ζ_{1i} vs. <i>COA</i> suggests homoscedasticity at both values of <i>COA</i> . So does a plot vs. <i>PEER</i> at least for values up to, and including, 2. Beyond this, there are too few cases to judge.

Checking Functional Form

(Singer and Willett, pg. 128)

- **At level-1.** For each individual, examine empirical growth plots and superimpose an OLS-estimated individual change trajectory. Inspection should confirm the suitability of its hypothesized shape.
- **At level-2.** Plot OLS estimates of the individual growth parameters against each level-2 predictor. Inspection should confirm the suitability of the hypothesized level-2 relationships.

Checking Functional Form

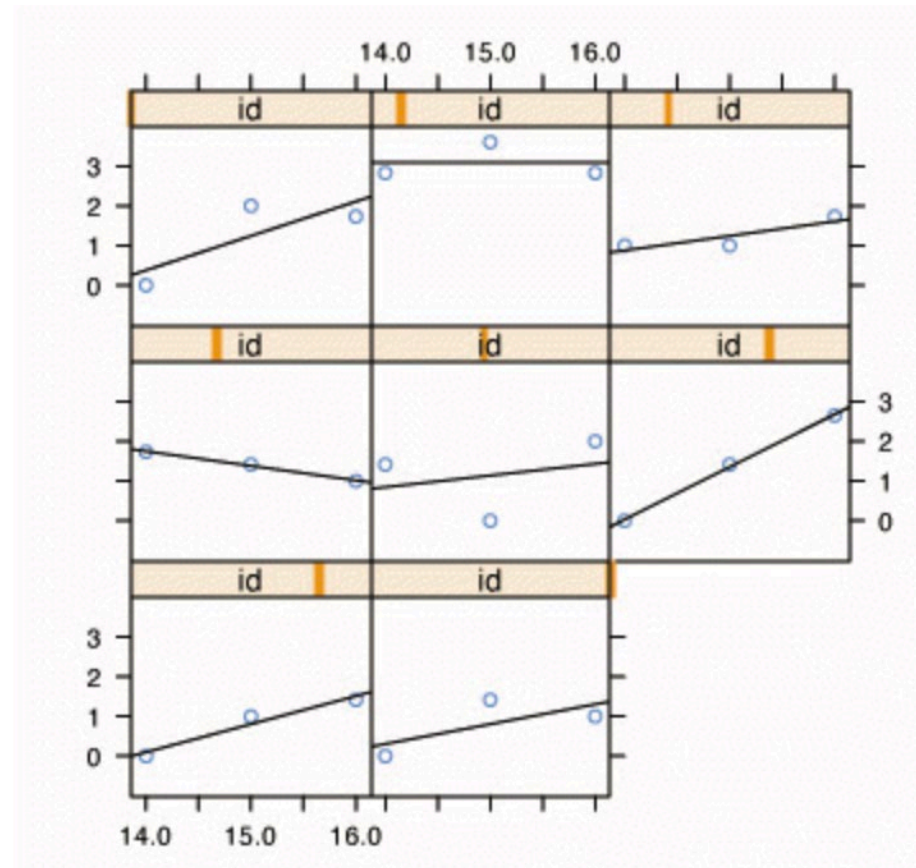
(Singer and Willett, pg. 128)

1

- Overall individual growth plots look linear to me.
 - You can take another random sample and see if you get the same overall pattern

```
library(lattice)

xyplot(alcuse~age | id,
  data=alcohol1[alcohol1$id %in% c(4, 14, 23, 32, 41, 56, 65, 82), ],
  panel=function(x,y){
    panel.xyplot(x, y)
    panel.lmline(x,y)
  }, ylim=c(-1, 4), as.table=T)
```

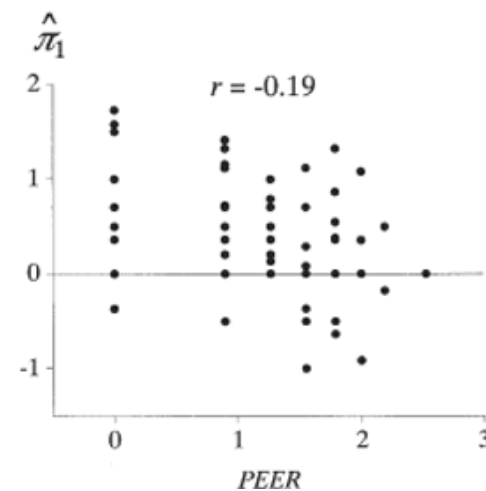
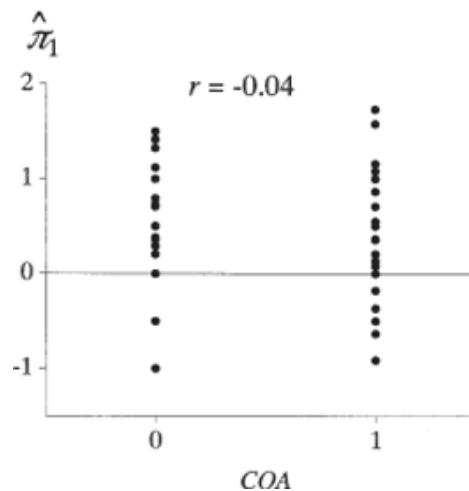
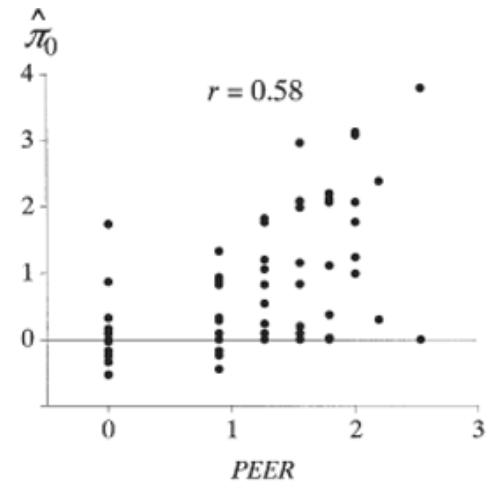
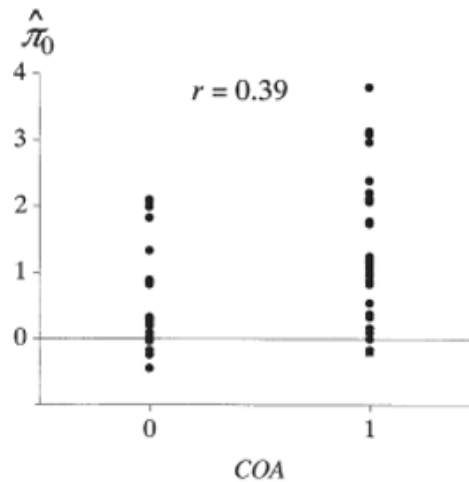


Checking Functional Form

(Singer and Willett, pg. 128)

#2

- Overall the OLS estimates look linear for each of the 2 time-invariant covariates
- When covariate dichotomous, nothing really to assess. You can see if there are any outliers
- The two plots for peers seem linear with some outliers (scatter about with no clear form)

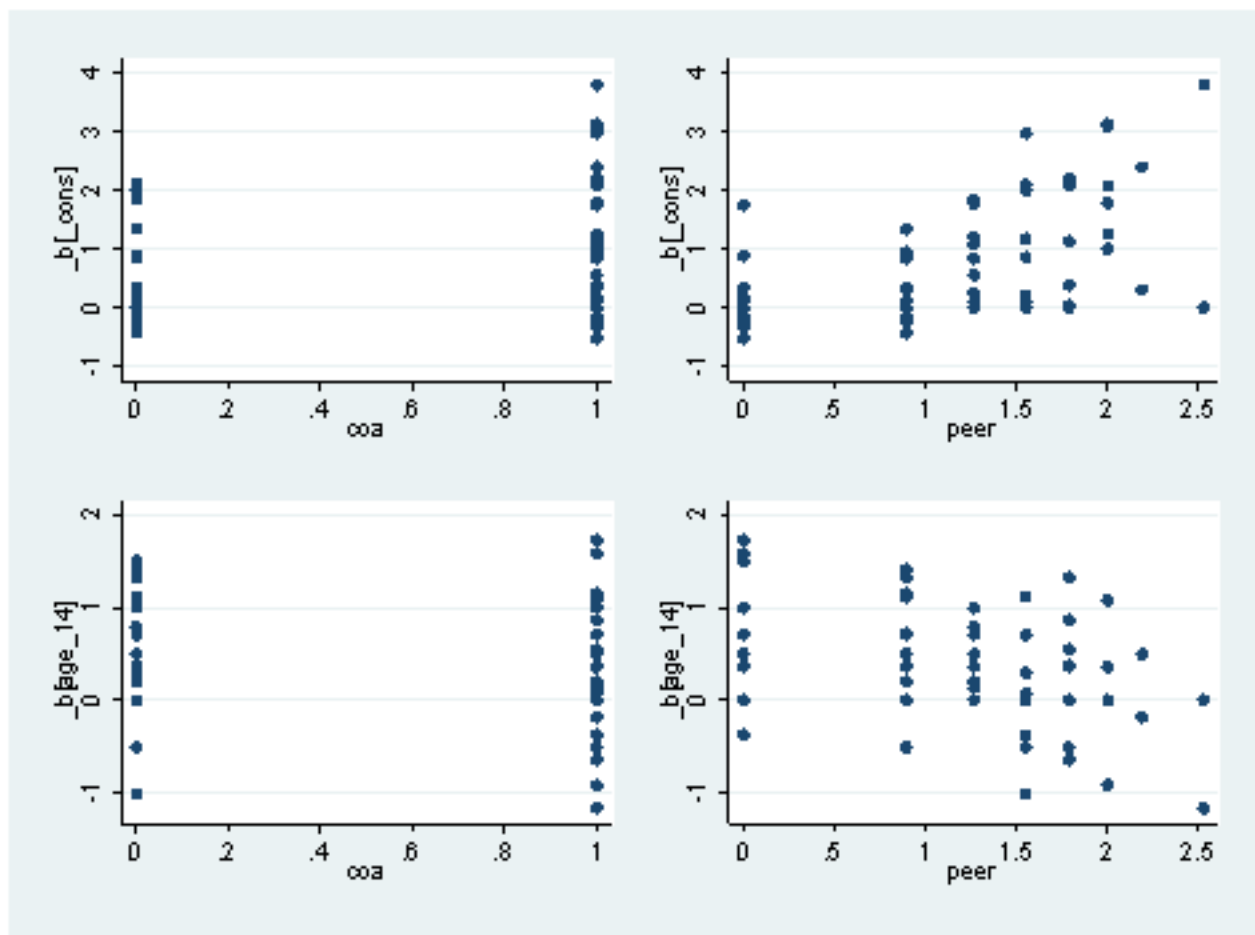


Checking Functional Form

(Singer and Willett, pg. 128)

#2

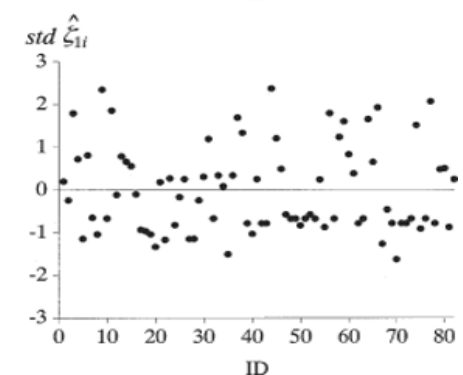
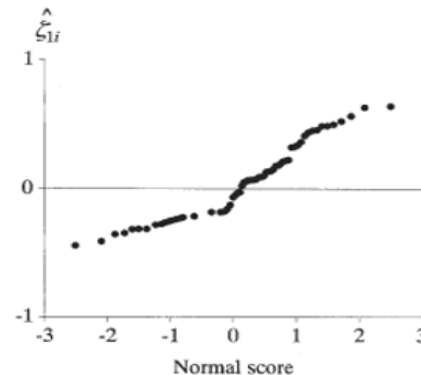
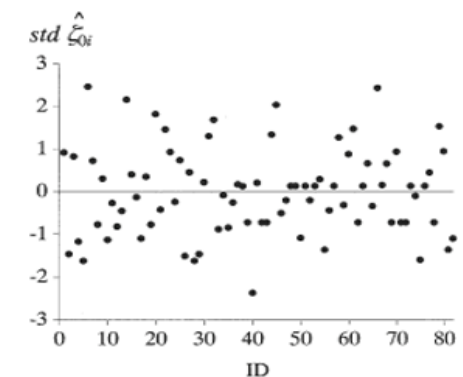
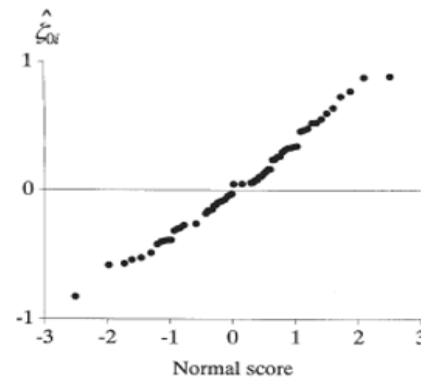
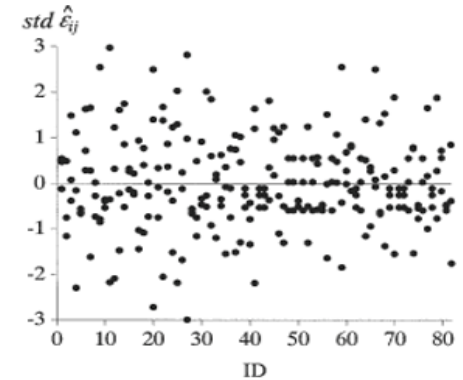
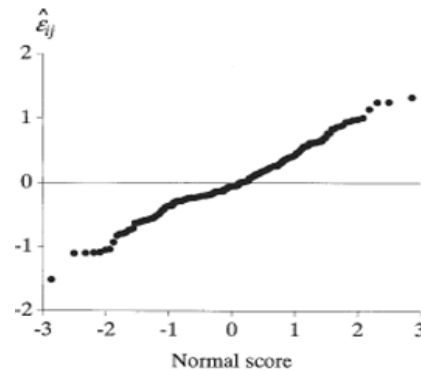
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Checking Normality Assumption

(Singer and Willett, pg. 130)

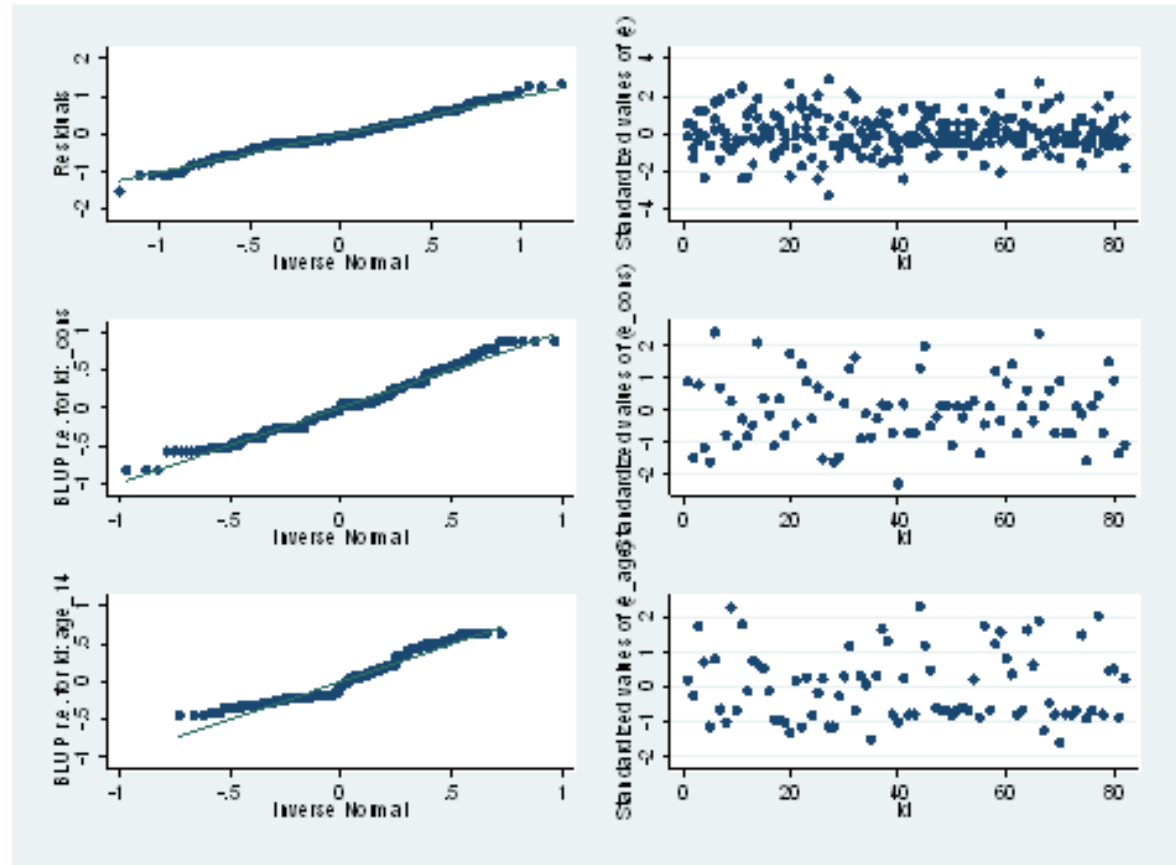
- Want to make sure the residuals are normally distributed
- Any departure from the line is a threat to normality.
- The std of the residuals should be scatter with no form around 0 and 95% of scatter within 2 stdv of 0
- It looks like, for **MODEL F**, the level-1 residuals are normal
- Level-2 residual may be concerning



Checking Normality Assumption

(Singer and Willett, pg. 130 [STATA version – see UCLA site])

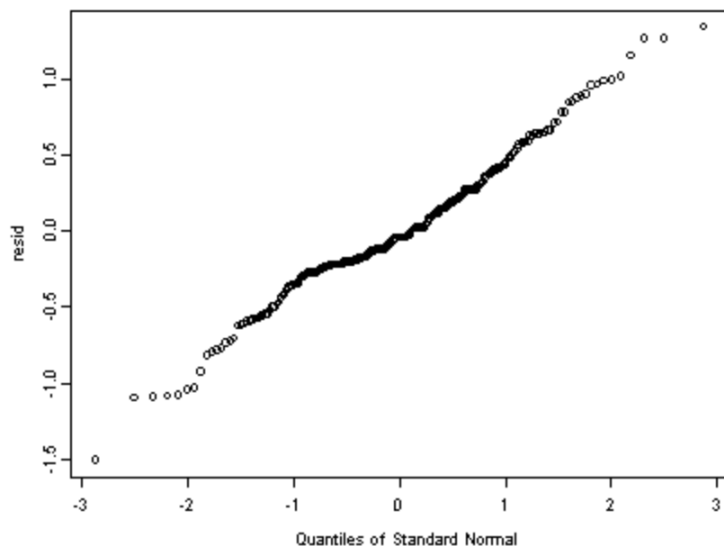
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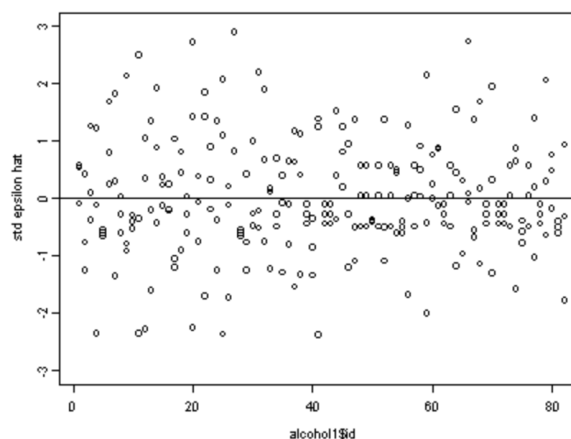
Checking Normality Assumption

(Singer and Willett, pg. 130 [R version – see UCLA site])

```
#creating the residuals (epsilon.hat)  
resid <- residuals(model.f)  
qqnorm(resid)
```



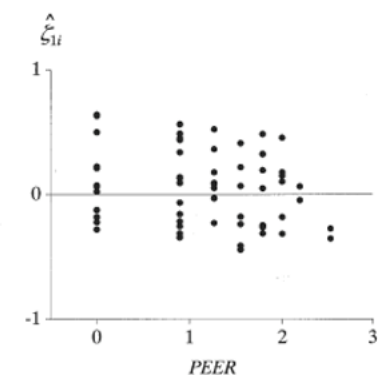
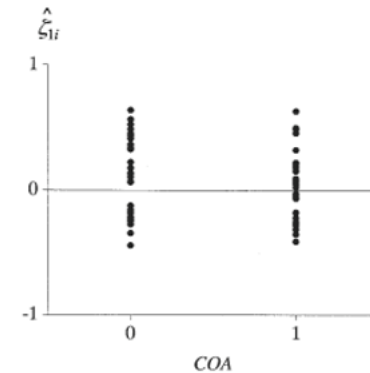
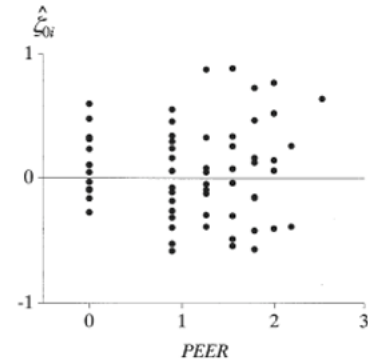
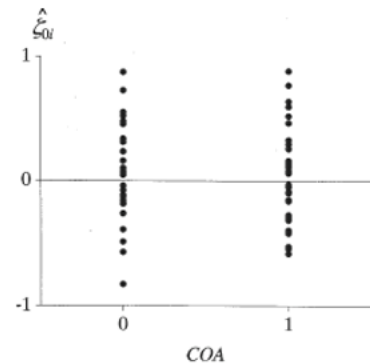
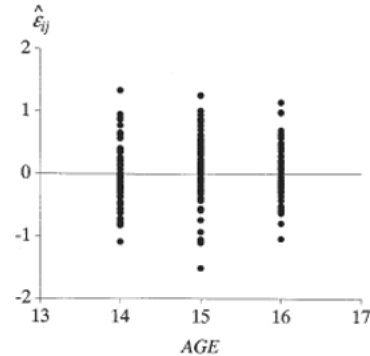
```
#creating the standardized residual (std epsilon.hat)  
resid.std <- resid/sd(resid)  
plot(alcohol1$id, resid.std, ylim=c(-3, 3), ylab="std epsilon hat")  
abline(h=0)
```



Checking Homoscedasticity

(Singer and Willett, pg. 133)

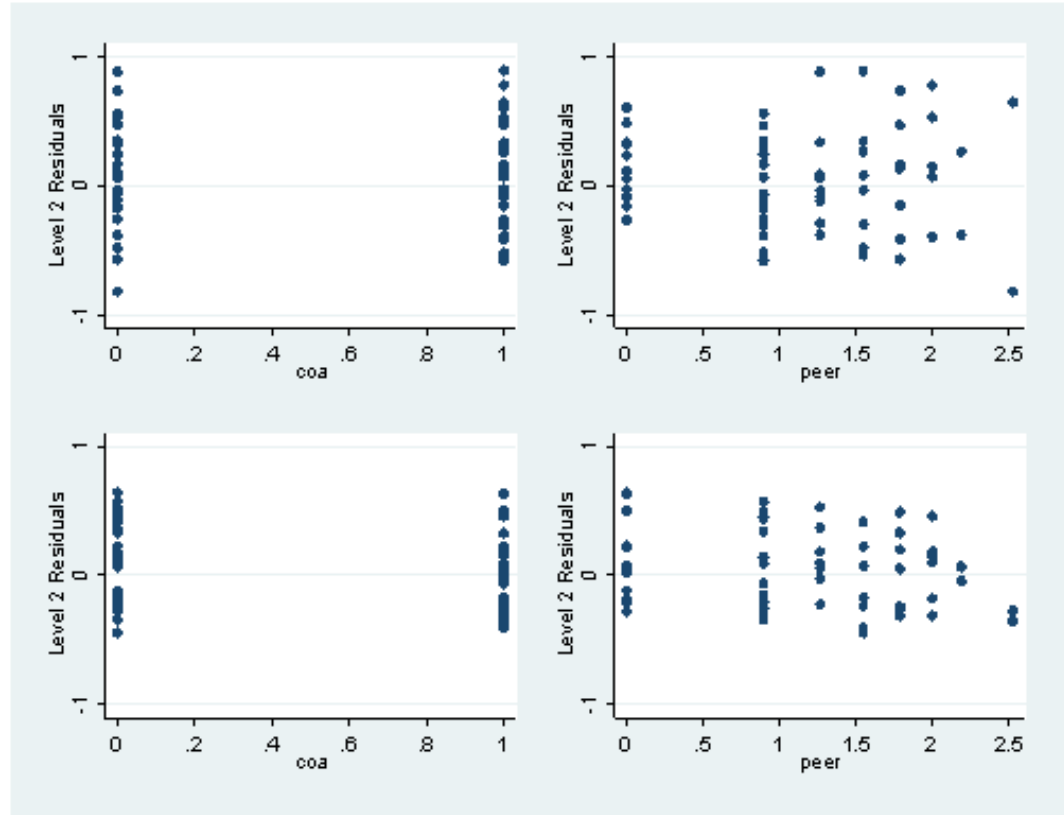
- Plotting the level-1 residuals against level-1 predictors
- Plotting the level-2 residuals against level-2 predictors
- If assumption holds the residual variance will be equal at each level of the predictor
- Level-1 and level-2 residuals seems to have equal range and variability at each time point
- At higher levels of PEERS there seems to be a drop in variability with predictor values greater than 2.5. This may suggest heterogeneity



Checking Homoscedasticity

(Singer and Willett, pg. 130 [STATA version – see UCLA site])

- Plotting the level-1 residuals against level-1 predictors
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Summary of Testing Assumptions

- Check for functional form, normality and homoscedasticity
- Our estimators can handle a bit of deviance in our assumptions, but you must do additional post hoc testing to make sure that you are receiving unbiased estimates
- When these assumptions are violated we may have to find alternative approaches
 - E.g. Non-linear model

Administrative Stuff

Final Paper Assignment

- See screenshare

Next Class...

- Estimators
- Time-varying covariates
 - **Singer and Willet pgs. 159- 181**
- Answer questions concerning assignment #2
- Discuss assignment #3

Class Exercises

Class Exercise 1: Assessing Model Fit and Testing Assumptions

1. Run the unconditional mean model, the unconditional growth model, and the “final” model with the covariate biological sex (may have to recode so it makes sense)
2. Determine the best model fit based on what you learned in today's lecture
3. Test the model assumptions for the “final” model

<https://stats.idre.ucla.edu/stata/examples/alda/chapter4/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-4-doing-data-analysis-with-the-multile/>

Class Exercise 2: Assessing Model Fit and Testing Assumptions

- Flint Adolescent Study
- Mean depression at year 1-5 (15-19 years of age)
- Sex (male; female)
- Directions same as "Class Exercise 1"